

Assignment No	:9
Roll No	:381
Batch	:E4
PRN No	:202502040032
Title of the Assignment	Java Server Faces

Problem Statement 1 : Write a program for JSF Standard Components and Facelets Tags.

Code:-

//UserBean.java

```
import javax.faces.bean.ManagedBean;
import javax.faces.bean.RequestScoped;
import java.util.Date;
```

```
@ManagedBean
@RequestScoped
public class UserBean {
```

```
    private String name;
    private int age;
    private Date dob;
```

```
    // Action method
    public String submit() {
        return "success";
    }
```

```
    // Getters and Setters
```

```
    public String getName() {
        return name;
    }
```

```
    public void setName(String name) {
        this.name = name;
    }
```

```
public int getAge() {  
    return age;  
}  
  
public void setAge(int age) {  
    this.age = age;  
}  
  
public Date getDob() {  
    return dob;  
}  
  
public void setDob(Date dob) {  
    this.dob = dob;  
}  
}
```

Index.xhtml:-

```
<html xmlns="http://www.w3.org/1999/xhtml"  
    xmlns:h="http://xmlns.jcp.org/jsf/html"  
    xmlns:f="http://xmlns.jcp.org/jsf/core">
```

```
<h:head>  
    <title>JSF Demo</title>
```

```
<style>  
    body {  
        font-family: Arial;  
        background-color: #f5f5f5;  
    }  
  
    .form-box {  
        width: 350px;  
        margin: 50px auto;  
        padding: 20px;  
        background: white;  
        border-radius: 8px;  
        box-shadow: 0px 0px 10px gray;  
    }
```

```
.error-box {
    color: #b30000;
    background-color: #ffe6e6;
    border: 1px solid red;
    padding: 10px;
    margin-bottom: 15px;
    border-radius: 5px;
}

.field-error {
    color: red;
    font-size: 12px;
}

.input-field {
    width: 100%;
    padding: 5px;
    margin-top: 5px;
}

.btn {
    margin-top: 15px;
    padding: 8px;
    width: 100%;
    background-color: #007bff;
    color: white;
    border: none;
    border-radius: 5px;
}

.output-box {
    margin-top: 15px;
    padding: 10px;
    background: #e6ffe6;
    border: 1px solid green;
    border-radius: 5px;
}
</style>
</h:head>

<h:body>
```

```
<h:form>
  <div class="form-box">

    <h2>User Form</h2>

    <!-- GLOBAL ERROR BOX -->
    <h:messages layout="table" styleClass="error-box" />

    <!-- Name -->
    <h:outputLabel value="Name:" />
    <h:inputText id="name"
      value="#{userBean.name}"
      required="true"
      requiredMessage="Name is required!"
      styleClass="input-field" />
    <h:message for="name" styleClass="field-error" />
    <br /><br />

    <!-- Age -->
    <h:outputLabel value="Age:" />
    <h:inputText id="age"
      value="#{userBean.age}"
      styleClass="input-field">
      <f:validateLongRange minimum="18"
        maximum="60"
        validatorMessage="Age must be between 18 and 60" />
    </h:inputText>
    <h:message for="age" styleClass="field-error" />
    <br /><br />

    <!-- DOB -->
    <h:outputLabel value="DOB:" />
    <h:inputText id="dob"
      value="#{userBean.dob}"
      styleClass="input-field">
      <f:convertDateTime pattern="dd-MM-yyyy"
        converterMessage="Enter date in DD-MM-YYYY format" />
    </h:inputText>
    <h:message for="dob" styleClass="field-error" />
    <br /><br />
```

```

        <!-- Submit Button -->
        <h:commandButton value="Submit"
            action="#{userBean.submit}"
            styleClass="btn" />

        <!-- OUTPUT DISPLAY -->
        <h:panelGroup rendered="#{not empty userBean.name}"
            styleClass="output-box">

            <h3>Entered Data:</h3>
            Name: #{userBean.name} <br />
            Age: #{userBean.age} <br />
            DOB: #{userBean.dob}
        </h:panelGroup>

    </div>
</h:form>

</h:body>
</html>

```

Web.xml:-

```

<web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
    version="3.1">

    <!-- JSF Faces Servlet Configuration -->
    <servlet>
        <servlet-name>Faces Servlet</servlet-name>
        <servlet-class>javax.faces.webapp.FacesServlet</servlet-class>
    </servlet>

    <!-- URL Mapping for JSF Pages -->
    <servlet-mapping>
        <servlet-name>Faces Servlet</servlet-name>
        <url-pattern>*.xhtml</url-pattern>
    </servlet-mapping>

</web-app>

```

Pom.xml:-

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    http://maven.apache.org/xsd/maven-4.0.0.xsd">

  <!-- Project Details -->
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>JSFProject</artifactId>
  <version>1.0</version>
  <packaging>war</packaging>

  <!-- Dependencies -->
  <dependencies>

    <!-- JSF API -->
    <dependency>
      <groupId>javax.faces</groupId>
      <artifactId>javax.faces-api</artifactId>
      <version>2.3</version>
    </dependency>

    <!-- JSF Implementation (Mojarra) -->
    <dependency>
      <groupId>org.glassfish</groupId>
      <artifactId>javax.faces</artifactId>
      <version>2.3.9</version>
    </dependency>

    <!-- Servlet API -->
    <dependency>
      <groupId>javax.servlet</groupId>
      <artifactId>javax.servlet-api</artifactId>
      <version>4.0.1</version>
      <scope>provided</scope>
    </dependency>

  </dependencies>

</project>
```

Output:-

User Form

Name is required!
j_idt5:age: Validation Error: Specified attribute is not between the expected values of 18 and 60.

Name:

Name is required!

Age:

j_idt5:age: Validation Error: Specified attribute is not between the expected values of 18 and 60.

DOB:

Submit

User Form

j_idt5:age: Validation Error: Specified attribute is not between the expected values of 18 and 60.

Name:

Age:

j_idt5:age: Validation Error: Specified attribute is not between the expected values of 18 and 60.

DOB:

Submit

User Form

Name:

Age:

DOB:

Submit

Name: sairaj

Age: 19

DOB: 28-12-2006

Submitted By:-

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